THE TRACE MAP FOR THE RIGHT ADJOINT BECOMES AN ISOMORPHISM AFTER APPLYING THE COHERATOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0A9Q (sources/stacks-duality/pinned-original-source.tex line 775).

Allowed standard prerequisites (editorial).

- Existence of the coherator and its adjointness to the inclusion. Derived Categories of Schemes, Lemma labelled perfect-lemma-better-coherator and Section labelled perfect-section-better-coherator, cited at proof lines 792-796 and in the statement. Makes the statement meaningful and reduces it to testing on D_QCoh.
- Adjunction for the pushforward along a quasi-compact quasi-separated morphism. Cohomology, Lemma labelled cohomology-lemma-adjoint, cited at proof lines 810-812. Gives the first and third equalities of the chain.
- The internal Hom adjunction. Cohomology, Lemma labelled cohomology-lemma-internal-hom, cited at proof lines 812-814 and 822-824. Gives the second and fifth equalities.
- The projection formula style base change for a tensor product. Derived Categories of Schemes, Lemma labelled perfect-lemma-cohomology-base-change, cited at proof lines 818-822. Gives the middle equality, marked as the important step.

Source reference. Original equation 0B6H: sources/stacks-duality/pinned-original-source.tex, source lines 758–758. Same-chapter pointer into the author environment equation at source line 758; invoked at proof line 798 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0CQZ: sources/stacks-chow/tags.txt, source lines 15979–15979. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is

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Source reference. Original lemma 0CR0: sources/stacks-chow/tags.txt, source lines 15980–15980. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 794 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 079W: sources/stacks-chow/tags.txt, source lines 9286–9286. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 818 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08DJ: sources/stacks-chow/tags.txt, source lines 10639–10639. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 819 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08EU: sources/stacks-chow/tags.txt, source lines 10685–10685. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 822 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of quasi-compact and quasi-separated schemes. Let a be the right adjoint to $Rf_* : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_Y)$. Let $L \in D_{QCoh}(\mathcal{O}_X)$ and $K \in D_{QCoh}(\mathcal{O}_Y)$. Then the map (0B6H)*

$$Rf_* R\mathcal{H}om_{\mathcal{O}_X}(L, a(K)) \longrightarrow R\mathcal{H}om_{\mathcal{O}_Y}(Rf_*L, K)$$

becomes an isomorphism after applying the functor $DQ_Y : D(\mathcal{O}_Y) \to D_{QCoh}(\mathcal{O}_Y)$ discussed in Derived Categories of Schemes, Section 0CQZ.

Proof. The statement makes sense as DQ_Y exists by Derived Categories of Schemes, Lemma 0CR0. Since DQ_Y is the right adjoint to the inclusion functor $D_{QCoh}(\mathcal{O}_Y) \to D(\mathcal{O}_Y)$ to prove the lemma we have to show that for any $M \in D_{QCoh}(\mathcal{O}_Y)$ the map (0B6H) induces an bijection

$$\mathrm{Hom}_Y(M, Rf_* R\mathcal{H}om_{\mathcal{O}_X}(L, a(K))) \longrightarrow \mathrm{Hom}_Y(M, R\mathcal{H}om_{\mathcal{O}_Y}(Rf_*L, K))$$

To see this we use the following string of equalities

$$\begin{aligned}
\mathrm{Hom}_Y(M, Rf_* R\mathcal{H}om_{\mathcal{O}_X}(L, a(K))) &= \mathrm{Hom}_X(Lf^*M, R\mathcal{H}om_{\mathcal{O}_X}(L, a(K))) \\
&= \mathrm{Hom}_X(Lf^*M \otimes^{\mathbf{L}}_{\mathcal{O}_X} L, a(K)) \\
&= \mathrm{Hom}_Y(Rf_*(Lf^*M \otimes^{\mathbf{L}}_{\mathcal{O}_X} L), K) \\
&= \mathrm{Hom}_Y(M \otimes^{\mathbf{L}}_{\mathcal{O}_Y} Rf_*L, K) \\
&= \mathrm{Hom}_Y(M, R\mathcal{H}om_{\mathcal{O}_Y}(Rf_*L, K))
\end{aligned}$$

The first equality holds by Cohomology, Lemma 079W. The second equality by Cohomology, Lemma 08DJ. The third equality by construction of a. The fourth equality by Derived Categories of Schemes, Lemma 08EU (this is the important step). The fifth by Cohomology, Lemma 08DJ. □